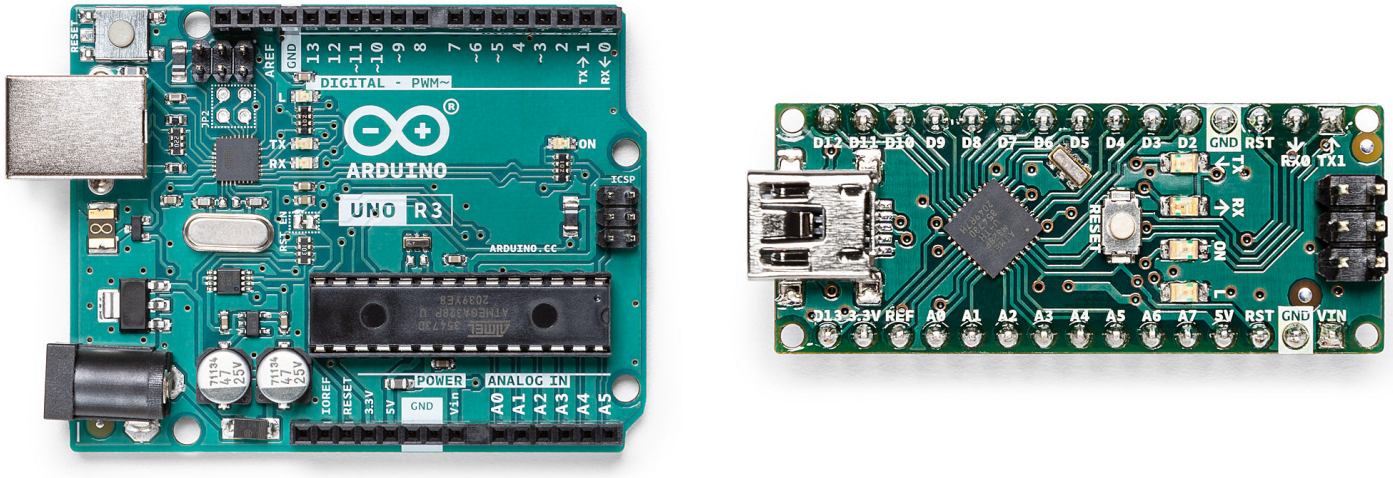
Chapter 1

Hardware

The Arduino platform is a flexible, relatively easy to learn, microprocessor capable of sensing, communication, and control. There are many different versions of the Arduino, but we will focus on the two most common and fitting versions for this class. These are shown in Figure 1.1.

You can think of the Arduino is a super simple computer, yet it's still different from the computer you probably use on a daily basis. The Arduino has a set of general purpose input and output pins (GPIO). This allows input and output of analog, digital, and communication protocols. Not all pins support all function though! See Figure 1.2 for details on the pins of the Uno. Don't worry about all of the acronyms in there yet though, as we'll go through those in the next few pages here.



(a) Arduino Uno R3, the most popular Arduino hardware.

(b) Arduino Nano, a much smaller format.

Figure 1.1: Arduino Hardware, which is constanly evolving. These two versions are likely to be continue to be available in some format in the near future.

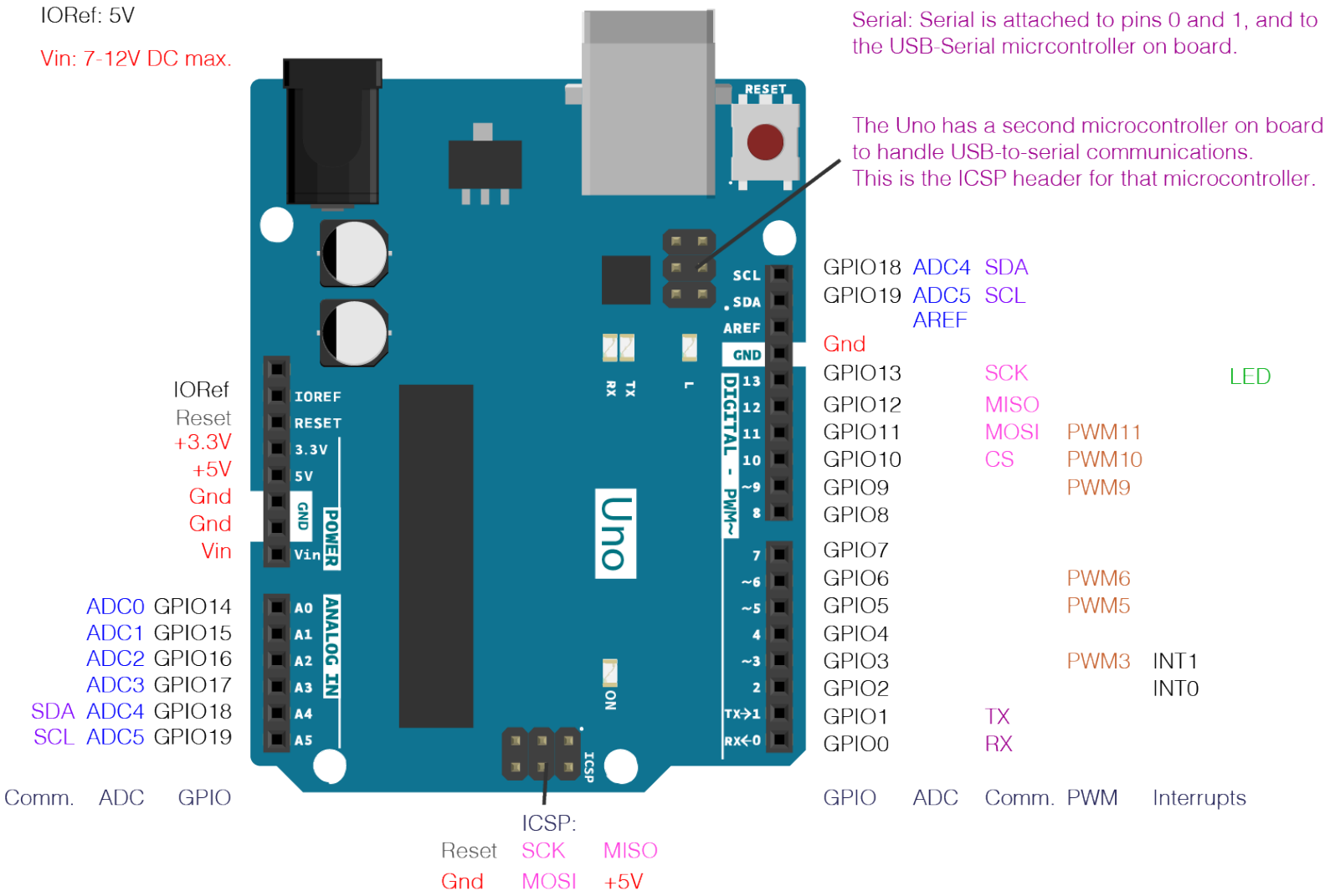


Figure 1.2: The Arduino Uno platform has 19 GPIO pins available for use. However, some of these pins support secondary functions. For example, you can see that A4 and A5 on the lower left are labeled as SDA ADC4 and SCL ADC5 respectively. SDA and SCL are communication type pins (discussed later). ADC are analog to digital converter pins (also discussed later). These pins cannot be used for both of these functions simultaneously.

Chapter 2

Software

2.1 Code Outline

There are three major parts to Arduino code layout. Let's start with the *second* section.

In this second section, as stated in the comments, code will run one time at start up. This is where we will put our code that defines how pins on the General Purpose Input/Output (GPIO) will behave. This could be setting a pin to be used as output, input, or even a communication interface.

In the *third* section we'll put the code that will run over and over again. This comes right after the setup() loop.

Finally, we need to declare variables that we want to be scoped globally. What does this mean? Any variables that we want to be able to access in the setup() and the loop(). If you're new to programming an Arduino, then it might be best to just assume that all variables should be global for now. There are indeed cases where you will want a variable to be accessible only inside of one of these, or a different function that you might write, but that's for later...

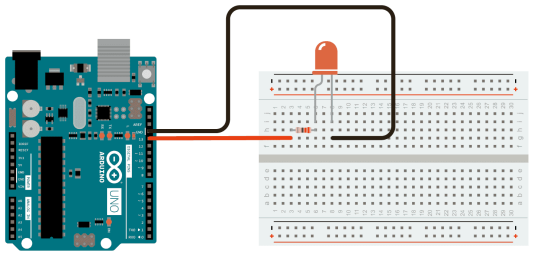
```
// put your variable declarations here.
// and your includes.

void setup() {
  // put your setup code here, to run once:

}

void loop() {
  // put your main code here, to run repeatedly:

}
```

2.2 Blinking a Light Emitting Diode (LED)

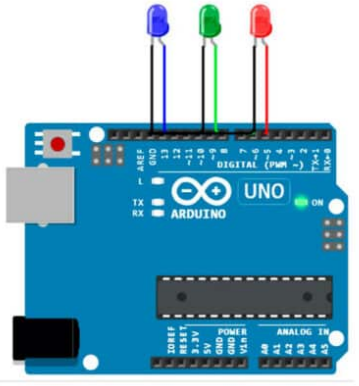
What about Ohm's Law?

We're going to make some assumptions for the moment, and will come back to this in **??**.

To start with, we'll get an LED connected to the Arduino to blink. Most Arduinos have a built in LED, but we're going to plug one in as a matter of practice. As shown here, we are using a breadboard and a resistor, in addition to the LED. However, in practice you can skip the breadboard and the resistor and connect your LED directly to the Arduino.



(a) Following best practice, we should have a resistor in our circuit.



(b) But we can also generally get away with being a little lazy and skipping the resistor. More on this later!

Figure 2.1: Arduino LED examples.

Here's the code; let's look at what's really going on here.

```
1  void setup() {
2    pinMode(13, OUTPUT); // Set pin 13 as an output
3  }
4
5  void loop() {
6    digitalWrite(13, HIGH); // Turn the LED on
7    delay(1000); // Wait for one second
8    digitalWrite(13, LOW); // Turn the LED off
9    delay(1000); // Wait for one second
10 }
```

First up, we are not using any variables to store state or values. We also are not using any special libraries, other supporting code, to do anything special. Thus, line 1 starts with our setup() call. Following this, we set the pin mode of GPIO pin 13 to be an output.

Next up, we have the loop(). These four lines turn the LED on by setting GPIO13 to "HIGH", pause for 1 second (1000ms), turn the LED off by setting GPIO13 to "LOW", pause for another 1 second, and then repeats. When we set a pin to "HIGH" we are setting it to "on", or thinking in binary, "TRUE" at 5 volts (assuming we are using a 5 volt Arduino). Likewise, setting the pin to "LOW", which you can also think of as "FALSE", sets the pin to "off".

Is there a state between "HIGH" and "LOW"?

Yes, but we need to save that for the section on pulse width modulation.

2.3 Adding a counter

In this course, we are using the Arduino as a platform for collecting data. Blinking an LED isn't really collecting data, or least not unless you want to sit by the LED and count how many times it blinks before the battery running the Arduino gives up. So let's introduce another function that is built into the Arduino, one that will let us communicate with the Arduino.

For now, we have been plugging our Arduino into our computer to upload our programs, and incedentially, to power the Arduino. With the introduction of this next function, we are assuming that is still the case.

What is baud?

Baud, or baud

```
1  int ledBlinks = 0;      // Create a variable to store the number of times the LED has \
       blinked for us
2
3  void setup() {
4    pinMode(13, OUTPUT); // Set pin 13 as an output
5    Serial.begin(9600);  // Define that we will be using the serial port, and 9600 baud
6  }
7
8  void loop() {
9    digitalWrite(13, HIGH); // Turn the LED on
10   delay(1000); // Wait for one second
11   digitalWrite(13, LOW); // Turn the LED off
12   delay(1000); // Wait for one second
13   ledBlinks = ledBlinks + 1;
14   Serial.print(ledBlinks);
15 }
```